

Botond Branyicskai-Nagy

London, United Kingdom

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Education

University College London

September 2023 – (September 2024)

MSc Machine Learning

Probabilistic and Unsupervised Learning¹ | Approximate Inference¹ | Applied ML

Bayesian Deep Learning | Supervised Learning | Reinforcement Learning²

Applied Deep Learning | ML Seminar

Imperial College London

2020 – 2023

BSc Physics with Theoretical Physics

Eotvos Jozsef Secondary School, Budapest

2014 – 2020

A-levels equivalent in Hungary at advanced level: Mathematics, Physics, English

Research Experience

MSc Thesis - University College London

2024

Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning: wake-sleep Bayesian algorithm learning with neural library of primitives, and a transformer-based generative model for synthesis. Supervised by Prof. Mirco Musolesi.

Research Internship — Imperial College London

2023

Causal discovery using cross-validation as compared with Bayesian Model Selection (GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP.

BSc Thesis — Imperial College London

2023

Computational Evolution: Simulation and analysis of evolution by tracking agents in virtual environments. Comparison with theoretical models, exploration of optimisation algorithms.

Other Projects

Masked Image Modelling — University College London

2024

Self-supervised learning framework: a VisionTransformer is pre-trained to reconstruct partially masked images then fine-tuned on the Oxford-IIIT Pets dataset for segmentation.

Barabasi-Albert and Oslo Models — Imperial College London

2023

Implementation of a simulation and analysis of each model's behaviour.

Solar Sail (Project Svarog) — Imperial College Space Society

2022

Orbital mechanics: spacecraft trajectory simulation, atmospheric effects, relativistic corrections.

Machine learning for LHCb experiment — Imperial College London

2022

Leading ML team in project analysing decay product distribution from LHCb experiment for Standard Model comparison. Neural Networks (TensorFlow), Decision Trees.

Laser microphone — Imperial College London

2020

Building a proof-of-concept prototype of a laser interferometer-based device capable of recovering audio from vibrations of a window from a distance.

Awards

2nd Year Quantum Physics Article Prize – Published on Imperial Website (2022)

Young Researchers of Natural Sciences, Conference Presentation Distinction (2016)

Skills and Interests

Languages & Programming

Python - Scientific and ML libraries (TensorFlow, GPflow, JAX, PyTorch), Julia, Cluster usage

Languages

Hungarian (Native), English (Proficient - Cambridge Advanced 206), French (Intermediate)

Other Interests: Space Society, tennis, hiking, piano

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind

For academic references please contact:

Dr Mark van der Wilk

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Department of Computer Science,

University of Oxford

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Professor Daniel Waldram

Professor of Theoretical Physics,

Head of the Theoretical Physics Group,

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